

Roll No.

TCE-505

B. TECH. (CIVIL ENGINEERING) (FIFTH SEMESTER)

MID SEMESTER EXAMINATION, Oct., 2022

STRUCTURAL ANALYSIS—II

Time : 1½ Hours

Maximum Marks : 50

- Note :** (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each sub-question carries 10 marks.

1. (a) A symmetrical two-hinged parabolic arch, hinged at the springing, has a horizontal span of 32 m with a central rise of 5 m. Find the horizontal thrust at the hinges, when the middle half of the span carries a load of 20 kN/m measured horizontally. Assume secant variation of the moment of inertia. 10 Marks (CO1)

OR

- (b) A two-hinged semi-circular arch of uniform cross-section has a radius of 8 m. It is subjected to a point load of 60 kN acting at a section lying in the left half span and subtending an angle 30 degrees with horizontal. Working from first principles, determine the horizontal thrust in the arch. 10 Marks (CO1)
2. (a) A suspension bridge of 100 m span has a three-hinged stiffening girder supported by cables having a central dip of 10 m. The left half of the

P. T. O.

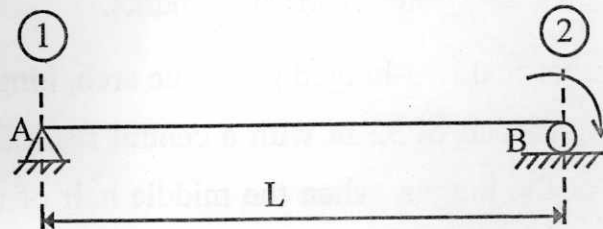
span is loaded with a uniformly distributed load of 30 kN/m. Determine the reactions and draw the bending moment diagram. 10 Marks (CO1)

OR

- (b) The stiffening girder of a suspension bridge of span 120 m has hinges at the ends and at mid-span. The cable is suspended between two spans separated horizontally by 120 m and vertically by 10 m. The maximum dip of the cable is 10 m below the lower suspension point. Calculate the maximum tension in the cable, due to concentrated load of 180 kN acting at the central hinge of stiffening girder. 10 Marks (CO1)

3. (a) Determine the flexibility matrix for the beam shown below :

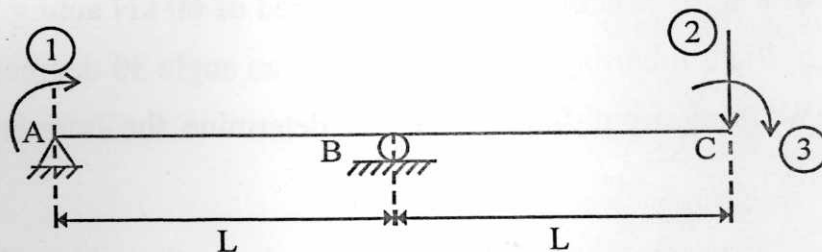
10 Marks (CO1)



OR

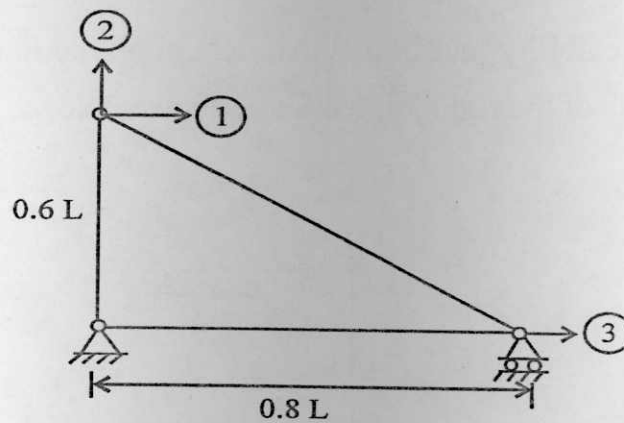
- (b) Determine the flexibility matrix for the beam shown below :

10 Marks (CO1)



4. (a) Determine the flexibility matrix for the truss shown below :

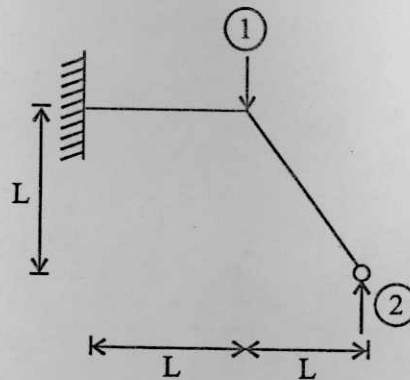
10 Marks (CO2)



OR

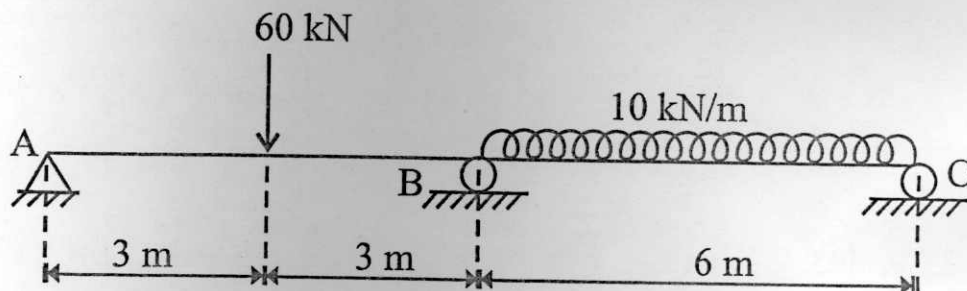
- (b) Determine the flexibility matrix for the frame shown below :

10 Marks (CO2)



5. (a) Using flexibility matrix method, determine the reactions of the beam shown below :

10 Marks (CO2)



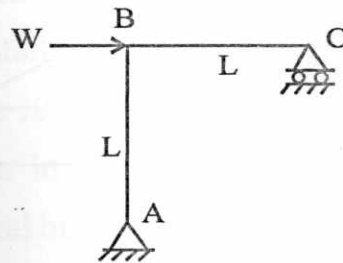
P. T. O.

(4)

TCE-505

OR

- (b) Using flexibility matrix method, determine the horizontal displacement of joint C of the rigid jointed frame shown below : 10 Marks (CO2)



TCE-505

90